10–15 第2篇

阅读理解（说明文） Why Do Fireflies Glow?

① On a warm summer night in the countryside, you may see **small green–yellow lights** moving slowly above the grass. They blink, disappear, and come back again, as if someone were writing tiny words in the air. These are **fireflies** — not small stars, not spirits, but little beetles with a very special gift: they can **make their own light**.

② A firefly does not burn like a candle. The light from a candle is mostly heat, and only a small part is light. But inside a firefly's body, almost **all the energy turns into light, and very little into heat**. That is why scientists call the firefly's light "**cold light**". If you hold a firefly gently in your hand, the light does not burn your skin. Even today, factories try to copy this kind of light, because it wastes almost no energy.

③ How does it work? In the firefly's **tail**, there is a chemical called **luciferin**. When luciferin meets oxygen, with the help of a special protein called luciferase, a reaction happens and light is produced. The firefly does **not** glow all the time; it controls the light by controlling how much air (and so how much oxygen) enters the tail. That is why the light blinks: **on, off, on, off**, like a tiny torch with a careful finger on the switch.

④ But the bigger question is: why do they bother? The answer is mainly **for love**. Male fireflies fly low above the grass, blinking in a pattern that belongs only to their own kind. Female fireflies wait in the grass and, if they like the pattern, blink back. In this way, the male and the female find each other in the dark. Every species has its own "light language". Some blink fast and short, some slow and long, and a few even draw J–shaped lines through the air.

⑤ Fireflies also use light to **warn enemies**. Their body contains a bitter chemical that most birds and spiders do not like. By glowing, the firefly seems to say, "I taste bad — please don't eat me." Young fireflies, which are too small to fly, can glow too, for the same reason. Nature, it seems, has turned their light into both **a love letter and a stop sign**.

⑥ Sadly, fireflies are getting harder to find. Bright street lamps, car lights and tall advertising boards make the night sky too bright, and fireflies can no longer read each other's signals in this "light pollution". Chemicals in the fields also kill their food — small snails and worms. In many Chinese cities, children today have never seen a real firefly.

⑦ The good news is that some villages in Jiangsu, Yunnan and Guangxi are now **turning off unnecessary lights on summer nights**, protecting rivers and grass, and setting up small "firefly parks". On a good night, you may still see a hundred tiny lights dancing above the water — a living reminder that **a dark sky is not empty; it is full of quiet life waiting to shine**.

注释： firefly 萤火虫；luciferin 荧光素；luciferase 荧光素酶；cold light 冷光；light pollution 光污染

根据短文内容，选择最佳答案。

1. What is the main difference between a firefly's light and a candle's light?

A. The firefly's light is stronger.
B. The firefly's light contains almost no heat, while most of a candle's energy turns into heat.
C. The firefly's light is always red.
D. A firefly's light is dangerous to touch.

2. How does the firefly control its blinking?

A. By closing its eyes.
B. By flapping its wings.
C. By controlling how much air (and so how much oxygen) enters its tail.
D. By jumping in the grass.

3. According to Paragraph ④, why do fireflies glow above the grass?

A. To warm up their bodies.
B. To guide humans.
C. To attract a mate — male and female find each other through different "light patterns".
D. To compete in a show.

4. What does the firefly's light tell its enemies, according to Paragraph ⑤?

A. "I am very fast."
B. "Follow me."
C. "I taste bitter — please don't eat me."
D. "I am a bird."

5. What does the writer's attitude seem to be in the last two paragraphs?

A. Angry and hopeless about the future of fireflies.
B. Not interested in fireflies at all.
C. Worried about light pollution but hopeful because some villages are protecting the dark sky for fireflies.
D. Satisfied that fireflies now live everywhere.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "almost all the energy turns into light, and very little into heat ... cold light"
【解析】萤火虫之光几乎不产热，烛光则主要为热。A、C、D 均无依据或错误。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段 "controls the light by controlling how much air (and so how much oxygen) enters the tail"
【解析】靠控气调光。A、B、D 均错。故选 C。

3.
【答案】C
【题型】细节理解题
【定位】第④段 "mainly for love ... male ... female wait ... light language"
【解析】为求偶而发光。A、B、D 均错。故选 C。

4.
【答案】C
【题型】细节理解题
【定位】第⑤段 "bitter chemical ... 'I taste bad — please don't eat me.'"
【解析】发光是化学警告。A、B、D 均错。故选 C。

5.
【答案】C
【题型】推理判断题（作者态度）
【定位】第⑥⑦段：忧光污染 + 赞村庄"萤火虫公园"
【解析】先忧后希望。A 否定态度太强，B 与全文相反，D 与事实相反。故选 C。